

**16.0.1 NIELIT 2017 Scientist B (CS) - Section A: 5**

The present ages of three persons in proportions $4 : 7 : 9$. Eight years ago, the sum of their ages was 56. Find their present ages (in years).

- A. 8, 20, 28
C. 20, 35, 45

- B. 16, 28, 36
D. None of the above options

nielit2017-scientistb-cs general-aptitude

Answer key

16.0.2 NIELIT 2017 Scientist B (CS) - Section A: 3

The percentage profit earned by selling an article for Rs.1,920 is equal to the percentage loss incurred by selling the same article for Rs.1,280. At what price should the article be sold to make 25% profit?

- A. Rs.2,000 B. Rs.2,200 C. Rs.2,400 D. Data inadequate

nielit2017-scientistb-cs general-aptitude

Answer key

16.0.3 NIELIT 2017 Scientist B (CS) - Section A: 1

What is the maximum number of distinct handshakes that can happen in the room with 5 people in it?

- A. 15 B. 10 C. 6 D. 5

nielit2017-scientistb-cs general-aptitude

Answer key

16.0.4 NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 17

A can is filled with 5 paise coins. Another can is filled with 10 paise coins. Another can is filled with 25 paise coins. All the cans are given wrong labels. If the can labeled 25 paise is not having the 10 paise coins, what will the can, labeled 10 paise have?

- A. 25 paise B. 5 paise
C. 10 paise D. cannot be determined

nielit2017oct-assistanta-cs general-aptitude quantitative-aptitude

Answer key

16.0.5 NIELIT Scientist B 2020 November: 28

Gopal went to a fruit market with certain amount of money. With this money he can buy either 50 oranges or 40 mangoes. He retains 10% of the money for taxi fare. If he buys 20 mangoes, then the number of oranges he can buy with balance amount is:

- A. 25 B. 20 C. 18 D. 6

nielit-scb-2020 quantitative-aptitude

Answer key 

16.0.6 NIELIT 2021 Dec Scientist B - Section A: 27



In a fraction, numerator is increased by 25% and the denominator is diminished by 10%. The new fraction obtained is $\frac{5}{9}$. The original fraction is :

- A. $\frac{2}{5}$ B. $\frac{5}{9}$ C. $\frac{3}{5}$ D. None of the above

nielit2021dec-scientistb quantitative-aptitude

Answer key 

16.0.7 NIELIT 2021 Dec Scientist B - Section A: 31



A, B and C rented a pasture by paying ₹ 2160 per month. They put 60, 40 and 20 sheep respectively. A sells $\frac{1}{3}$ of his sheep to B after 6 months and after 3 months more C sells $\frac{2}{5}$ of his sheep to A. Find the rent paid by C at the end of the year :

- A. ₹ 4355 B. ₹ 3888
C. ₹ 2464 D. ₹ 6224

nielit2021dec-scientistb quantitative-aptitude analytical-aptitude

16.0.8 NIELIT 2022 April Scientist B | Section A | Question: 20



Today it is Thursday. After 132 days, it will be :

- A. Monday B. Sunday C. Wednesday D. Thursday

nielit2022apr-scientistb analytical-aptitude quantitative-aptitude

Answer key 

16.0.9 NIELIT 2022 April Scientist B | Section A | Question: 15



Which largest number of five digits is divisible by 99?

- A. 99999 B. 99981 C. 99909 D. 99990

nielit2022apr-scientistb quantitative-aptitude

Answer key 

16.0.10 NIELIT 2023 Feb Scientist D | Section C | Question: 77



Find the smallest number that must be subtracted from 9100 to make it perfect

square?

- A. 25 B. 50 C. 75 D. 100

nielit2023feb-scientistd quantitative-aptitude

16.0.11 NIELIT 2023 Feb Scientist D | Section C | Question: 72



Find the least number that must be added to 7900 to obtain a perfect square.

- A. 67 B. 87 C. 97 D. 77

nielit2023feb-scientistd quantitative-aptitude

16.0.12 NIELIT 2023 Feb Scientist D | Section C | Question: 69



In a business, B is a sleeping partner and A is a working partner. A invests 5000 and B invests 6000. A receives 12.5% of profit for managing the business and the remaining amount is divided in proportion to their capitals. A's share of profit in a profit of Rs. 880 is :

- A. 350 B. 400 C. 420 D. 460

nielit2023feb-scientistd quantitative-aptitude

Answer key

16.0.13 NIELIT 2023 Feb Scientist D | Section C | Question: 66



In a Km race A can beat B by 80 m and B can beat C by 60 m. In the same race, A can beat C by _____.

- A. 135.2 m B. 130.5 m
C. 142 m D. 132.5 m

nielit2023feb-scientistd quantitative-aptitude analytical-aptitude

16.0.14 NIELIT 2023 Feb Scientist D | Section D | Question: 119



Ram visits recreation club after every five days, Harish visits after every 24 days, while Rajesh goes there after every 9 days. If all three of them met at the club on a Sunday. The days of the week will all three meet again is :

- A. Sunday B. Thursday C. Wednesday D. Monday

16.0.15 NIELIT 2023 Feb Scientist D | Section D | Question: 118

From a rope of length $38\frac{3}{5}$ m, a piece of length $5\frac{3}{38}$ m is cut off. The length of remaining rope is:

- A. $190\frac{99}{33}$ B. $33\frac{90}{190}$ C. $33\frac{99}{190}$ D. 1

16.0.16 NIELIT 2023 Feb Scientist C | Section C | Question: 89

Find the least number that when divided by 16, 18 and 20 leaves a remainder 4 in each case, but is completely divisible by 7.

- A. 364 B. 2254 C. 2964 D. 2884

16.0.17 NIELIT 2023 Feb Scientist C | Section C | Question: 87

Ramesh has total of Rs 56000 in four denominations of currency notes namely Rs 100, Rs 200, Rs 500, and Rs 2,000. How many total currency notes he has, if equal number of currency notes in each denomination are held by him.

- A. 28 B. 32 C. 24 D. 20

16.0.18 NIELIT 2023 Feb Scientist C | Section C | Question: 71

Which is the largest of the following fraction?

- A. $\frac{3}{5}$ B. $\frac{2}{3}$ C. $\frac{8}{11}$ D. $\frac{11}{17}$

Answer key

16.0.19 NIELIT 2023 Feb Scientist C | Section D | Question: 120

A man invested $\frac{1}{3}$ rd of his capital at 7%, $\frac{1}{4}$ th at 8% and remainder at 10%. If his annual income is Rs. 561. Find the capital.

A. Rs. 5000

B. Rs. 6000

C. Rs. 6600

D. Rs. 5600

nielit2023feb-scientistc quantitative-aptitude

16.0.20 NIELIT 2023 Feb Scientist C | Section D | Question: 117



Ram and shyam are two different persons, but when they worked together, they completed a job in 10 days. Had Ram worked at half of his efficiency and shyam at five times of his efficiency, they would have finished the job in 50% of the scheduled time. Shyam can complete the job working alone in y days. Then y is :

A. 20 days

B. 30 days

C. 40 days

D. 35 days

nielit2023feb-scientistc quantitative-aptitude analytical-aptitude

16.0.21 NIELIT 2023 Feb Scientist C | Section D | Question: 116



The number of integral pair (x, y) satisfying the equation: $(x - y)^2 + 2y^2 = 27$ is :

A. 8

B. 7

C. 6

D. 5

nielit2023feb-scientistc analytical-aptitude

16.0.22 NIELIT 2023 Feb Scientist C | Section D | Question: 115



A sum of money was invested at simple interest at a certain rate for 3 years. Had it been invested at a 4% higher rate it would have fetched Rs.480 more. Find the principal.

A. Rs. 4000

B. Rs. 5000

C. Rs. 6000

D. Rs. 3000

nielit2023feb-scientistc quantitative-aptitude analytical-aptitude

16.0.23 NIELIT 2023 Feb Scientist C | Section D | Question: 114



A shopkeeper advertises for selling cloth at 4% loss. However, by using a false meter scale he actually gains 25%. What is the actual length of the scale?

A. 76.8 cm

B. 77.8 cm

C. 74.8 cm

D. 75.8 cm

nielit2023feb-scientistc quantitative-aptitude analytical-aptitude

16.0.24 NIELIT 2023 Feb Scientist C | Section D | Question: 102



P, Q, R are three typist who working simultaneously can type 216 pages in 4 hours. In one hour, R can type as many pages more than Q as Q can type more than P. During a period of 5 hours, R can type as many pages as P can during 7 hours. How many pages does each of them type per hour.

- A. 14,17,20 B. 15,17,22 C. 15,18,21 D. 16,18,22

nielit2023feb-scientistc quantitative-aptitude analytical-aptitude

16.0.25 NIELIT 2023 Feb Scientist C | Section D | Question: 101



x, y, z are three positive integers such that x is greater than y & y is greater than z then which of the following is always true:

- A. $y\%$ of z is less than $x\%$ of y . B. $y\%$ of x is less than $z\%$ of y .
C. $z\%$ of x is less than $y\%$ of z . D. All of the above.

nielit2023feb-scientistc analytical-aptitude quantitative-aptitude

16.1

Age Relation (1)

16.1.1 Age Relation: NIELIT 2018-1



A Professor passed one sixth of his life in childhood, one twelfth in youth, and one seventh more as a bachelor, five years after his marriage a son was born who died four years before his father at half his final age, then what is the age of Professor?

- A. 84 years B. 74 years C. 64 years D. 54 years

nielit-2018 general-aptitude quantitative-aptitude age-relation

Answer key

16.2

Algebra (1)

16.2.1 Algebra: NIELIT 2023 Feb Scientist D | Section D | Question: 105



If $x = 3 + 3^{\frac{2}{3}} + 3^{\frac{1}{3}}$, then the value of $x^3 - 9x^2 + 18x - 12$ is :

- A. 1 B. -1 C. 0 D. 2

nielit2023feb-scientistd algebra functions

Answer key

16.3

Alligation Mixture (1)

A. 30

B. 32

C. 35

D. 33

Answer key

A. $\frac{1}{8}$

B. $\frac{3}{4}$

C. $\frac{3}{8}$

D. $\frac{2}{5}$

Answer key 

A. $x + 2$

B. $x + 3$

C. $x + 1$

D. 1

5.6

A. 4:1

B. 9 : 1

C. 1:1

D. 17:3

Answer key 

16.6.2 Arithmetic Series: NIELIT 2023 Feb Scientist C | Section C | Question: 88

Find the sum of all two-digit numbers that give a remainder of 3 when they are divided by 7.



- A. 686 B. 676 C. 666 D. 656

nielit2023feb-scientistc arithmetic-series quantitative-aptitude

16.6.3 Arithmetic Series: NIELIT 2023 Feb Scientist C | Section D | Question: 107

Find the sum of all three digit numbers that gives a remainder 4 when they are divided by 5.



- A. 98270 B. 99270 C. 102090 D. 90270

nielit2023feb-scientistc quantitative-aptitude arithmetic-series

16.6.4 Arithmetic Series: NIELIT 2023 Feb Scientist C | Section D | Question: 108

The value of $5 \times 2 - [3 - \{5 - (7 + 2 \times 4 - 19)\}]$ is :



- A. 0 B. 14 C. 16 D. -5

nielit2023feb-scientistc quantitative-aptitude arithmetic-series

Answer key

16.7

Average (5)

16.7.1 Average: NIELIT 2022 April Scientist B | Section A | Question: 40

What is the average of first five multiples of 12 ?



- A. 36 B. 38 C. 40 D. 42

nielit2022apr-scientistb quantitative-aptitude array average

Answer key

16.7.2 Average: NIELIT 2023 Feb Scientist C | Section D | Question: 99

In the month of July of a certain year, the average daily expenditure of an organization was Rs. 68. For the first 15 days of the month, the average daily expenditure was Rs. 85 and for the last 17 days, Rs. 51. Find the amount spent by the organization on the 15th of the month.



A. Rs. 42

B. Rs. 36

C. Rs. 34

D. Rs. 52

nielit2023feb-scientistc average quantitative-aptitude

16.7.3 Average: NIELIT 2023 Feb Scientist D | Section C | Question: 62

It rained as much on Wednesday as on all the others days of the week combined. If the average rainfall for the whole week was 3 cm, then how much did it rain on Wednesday?

A. 2.625 cm

B. 3 cm

C. 10.5 cm

D. 15 cm

nielit2023feb-scientistd analytical-aptitude average quantitative-aptitude

Answer key**16.7.4 Average: NIELIT 2023 Feb Scientist D | Section C | Question: 89**

The average of 100 numbers is 60. Numbers 70 & 80 are removed. What is the new average?

A. 59.69

B. 50.69

C. 11.69

D. 27.27

nielit2023feb-scientistd quantitative-aptitude average

Answer key**16.7.5 Average: NIELIT 2023 Feb Scientist D | Section D | Question: 117**

A person travels three equal distances at a speed of x km/h, y km/h, z km/h respectively. What will be the average speed during the whole journey?

A. $\frac{xyz}{xy+yz+zx}$ B. $\frac{xy+yz+zx}{xyz}$ C. $\frac{3xyz}{xy+yz+zx}$ D. $\frac{3xyz}{x+y+z}$

nielit2023feb-scientistd average speed-time-distance quantitative-aptitude analytical-aptitude

16.8**Clock Time (1)****16.8.1 Clock Time: NIELIT 2022 April Scientist B | Section A | Question: 22**

A clock is set at 4 am. It loses 16 minutes in 24 hours. What will be the correct time when the clock indicates 9 pm on the 4th day?

A. 8 pm

B. 7 pm

C. 10 pm

D. 11 pm

nielit2022apr-scientistb clock-time analytical-aptitude quantitative-aptitude

16.9

Compound Interest (4)

16.9.1 Compound Interest: NIELIT 2022 April Scientist B | Section A | Question: 32

What is the compound interest on Rs.2500 for 2 years at rate of interest 4% per annum ?



- A. Rs.180 B. Rs.204 C. Rs.210 D. Rs.220

nielit2022apr-scientistb quantitative-aptitude compound-interest

16.9.2 Compound Interest: NIELIT 2023 Feb Scientist C | Section C | Question: 65

A mother divided an amount of ₹61,000 between her two daughters aged 18 years and 16 years respectively and deposited their shares in a bond. If the interest rate is 20% compounded annually and if each received the same amount as the other when she attained the age of 20 years, their shares are :



- A. ₹35,600 and ₹25,400 B. ₹ 30500 each
C. ₹ 24,000 and ₹3700 D. None of these

nielit2023feb-scientistc quantitative-aptitude compound-interest

16.9.3 Compound Interest: NIELIT 2023 Feb Scientist D | Section D | Question: 100

Mr. Kishan invested money in two schemes A&B offering compound interest 8% per annum and 9% per annum respectively. If the total amount of interest accrued through two schemes together in two years was Rs. 4818.30 and the total amount invested was Rs. 27000. What was the amount invested in scheme A?



- A. Rs. 10000/— B. Rs. 13500/—
C. Rs. 15000/— D. Rs. 12000/—

nielit2023feb-scientistd quantitative-aptitude compound-interest

16.9.4 Compound Interest: NIELIT 2023 Feb Scientist D | Section D | Question: 120

A sum of money doubles itself at compound interest in 15 years. In how many years will it become eight times?



- A. 35 years B. 45 years C. 55 years D. 40 years

nielit2023feb-scientistd quantitative-aptitude compound-interest

16.10

Counting (1)

16.10.1 Counting: NIELIT Scientist B 2020 November: 6



A solid cube of each side 8 cm , has been painted red, blue and black on pairs of opposite faces. It is then cut into cubical blocks of each side 2 cm . How many cubes have no face painted?

- A. 0 B. 4 C. 8 D. 12

nielit-scb-2020 counting combinatory analytical-aptitude

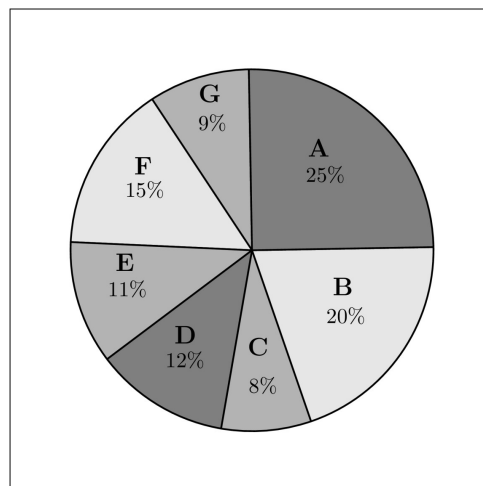
Answer key

16.11 Data Interpretation (2)

16.11.1 Data Interpretation: NIELIT Scientist B 2020 November: 1



Study the following graph and the table and answer the question given below.
(Data of different states regarding population of states in the year 2018)



Total population of the given States = 3276000

States	Sex and Literacy wise Population Ratio			
	Sex		Literacy	
	M	F	Educated	Non-Educated
A	5	3	2	7
B	3	1	1	4
C	2	3	2	1
D	3	5	3	2
E	3	4	4	1
F	3	2	7	2
G	3	4	9	4

_____ is the total number of non-educated people in *A* and *B* in 2018.

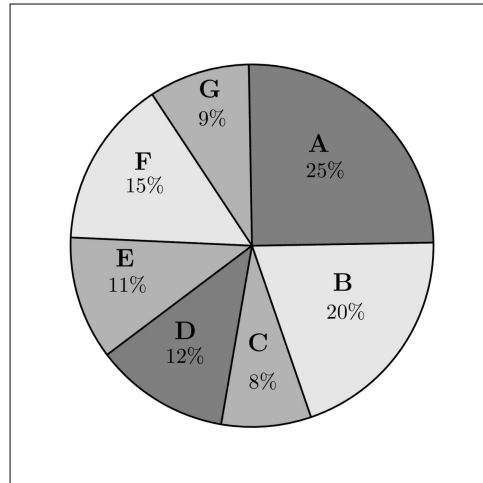
- A. 1276040 B. 1032170 C. 1081550 D. 1161160

Answer key

16.11.2 Data Interpretation: NIELIT Scientist B 2020 November: 2



Study the following graph and the table and answer the question given below.
(Data of different states regarding population of states in the year 2018)



Total population of the given States = 3276000

Sex and Literacy wise Population Ratio

States

	Sex		Literacy	
	M	F	Educated	Non-Educated
A	5	3	2	7
B	3	1	1	4
C	2	3	2	1
D	3	5	3	2
E	3	4	4	1
F	3	2	7	2
G	3	4	9	4

The number of males in *F* in the year 2018 is _____.

- A. 294650 B. 294840 C. 301470 D. 301200

Answer key

16.12.1 Decimal: NIELIT 2023 Feb Scientist D | Section D | Question: 107

How many digits will be there to the right of the decimal point in the product of 95.75 & 0.02554

- A. 5 B. 6 C. 7 D. 8

nielit2023feb-scientistd decimal quantitative-aptitude

16.13**Digit Sum (1)****16.13.1 Digit Sum: NIELIT 2017 DEC Scientist B - Section B: 53**

When the sum of all possible two digit numbers formed from three different one digit natural numbers are divided by sum of the original three numbers, the result is

- A. 26 B. 24 C. 20 D. 22

nielit2017dec-scientistb general-aptitude quantitative-aptitude digit-sum

Answer key 

16.14**Divisibility (2)****16.14.1 Divisibility: NIELIT 2023 Feb Scientist C | Section D | Question: 94**

Statement (I): 579876543 is divisible by 999

Statement (II): 6800900292 is divisible by 999

- A. Both (I) and (II) are true B. Both (I) and (II) are false
C. (I) is true and (II) is false D. (I) is false and (II) is true

nielit2023feb-scientistc number-theory divisibility analytical-aptitude quantitative-aptitude

16.14.2 Divisibility: NIELIT 2023 Feb Scientist D | Section D | Question: 97

Statement (I): 579876543 is divisible by 111

Statement (II): 6800900181 is divisible by 111

- A. Both (I) and (II) are true
B. Both (I) and (II) are false
C. (I) is true and (II) is false
D. (I) is false and (II) is true

nielit2023feb-scientistd divisibility number-theory analytical-aptitude

16.15**Factors (1)**

16.15.1 Factors: NIELIT 2023 Feb Scientist C | Section C | Question: 81

Two friends were discussing their marks in an examination. While doing so they realized that both the numbers had the same prime factors, although Raveesh got a score which had two more factors than Harish. If their marks are represented by one of the options as given below, which of the following options would correctly represent the number of marks they got?

- A. 30,60 B. 20,80 C. 40,80 D. 20,60

nielit2023feb-scientistc number-theory factors analytical-aptitude

16.16**Geometry (12)****16.16.1 Geometry: NIELIT 2021 Dec Scientist A - Section A: 28**

An aeroplane at an altitude of 3000 m observes the angles of depression of opposite points on the two banks of a river to be 45° and 60° respectively. Find the width of the river in metre.

- A. 4730 B. 4430 C. 4150 D. 4650

nielit2021dec-scientista geometry analytical-aptitude

16.16.2 Geometry: NIELIT 2021 Dec Scientist A - Section A: 38

The length, breadth and height of a cuboid are in the ratio $3 : 4 : 5$ and its volume is 3840 cm^3 , The smallest side has a length of :

- A. 12 cm B. 20 cm C. 15 cm D. 18 cm

nielit2021dec-scientista quantitative-aptitude geometry

Answer key

16.16.3 Geometry: NIELIT 2021 Dec Scientist B - Section A: 25

Instead of walking along two adjacent sides of a rectangular field, a boy took a short cut along the diagonal and saved a distance equal to half the longer side. Then, the ratio of the shorter side to the longer side is :

- A. $1/2$ B. $2/3$ C. $1/4$ D. $3/4$

nielit2021dec-scientistb quantitative-aptitude geometry

Answer key

16.16.4 Geometry: NIELIT 2021 Dec Scientist B - Section A: 30

A cylindrical vessel 60 cm in diameter is partially filled with water. A sphere 30 cm in a diameter is dropped into it. The increase in the level of water in the vessel is :

A. 2 cm

B. 3 cm

C. 4 cm

D. 5 cm

nielit2021dec-scientistb geometry quantitative-aptitude

Answer key 

16.16.5 Geometry: NIELIT 2021 Dec Scientist B - Section A: 32



A cuboid of dimension $24 \text{ cm} \times 9 \text{ cm} \times 8 \text{ cm}$ is melted and smaller cubes of side 3 cm are formed. Find how many such cubes can be formed :

A. 27

B. 64

C. 54

D. 32

nielit2021dec-scientistb quantitative-aptitude geometry

Answer key 

16.16.6 Geometry: NIELIT 2022 April Scientist B | Section A | Question: 1



A water tank is 30 m long, 20 m wide and 12 m deep. It is made of iron sheet which is 3 m wide. The tank is open at the top. If the cost of iron sheet is Rs. 10 per meter, what is the total cost of iron sheet required to build the tank?

A. Rs. 6000

B. Rs. 8000

C. Rs. 9000

D. Rs. 10,000

nielit2022apr-scientistb quantitative-aptitude geometry

Answer key 

16.16.7 Geometry: NIELIT 2022 April Scientist B | Section A | Question: 28



The tops of two poles are connected by a wire. The heights of the poles are 10 m and 14 m respectively. If the wire makes a 30° angle with the horizontal, find the length of the wire ?

A. 7 m

B. 7.5 m

C. 8 m

D. 9.5 m

nielit2022apr-scientistb geometry analytical-aptitude

16.16.8 Geometry: NIELIT 2023 Feb Scientist C | Section D | Question: 109



Seven equal cubes each of the side 5 cm are joined end to end. Find the surface area of the resultant cuboid.

A. 750 cm^2

B. 1500 cm^2

C. 2250 cm^2

D. 700 cm^2

nielit2023feb-scientistc geometry quantitative-aptitude

16.16.9 Geometry: NIELIT 2023 Feb Scientist D | Section C | Question: 76



The perimeter of a square plot is the same as that of a rectangular plot with sides 35 m and 15 m . The side of the square plot is:

A. 25 m

B. 20 m

C. 100 m

D. 50 m

nielit2023feb-scientistd geometry quantitative-aptitude

Answer key 

16.16.10 Geometry: NIELIT 2023 Feb Scientist D | Section D | Question: 106



The area of similar triangles ABC & DEF are 144 cm^2 and 81 cm^2 respectively. If the longest side of larger triangle ABC be 36 cm then the longest side of smaller triangle DEF is :

- A. 20 cm
C. 27 cm

- B. 26 cm
D. 30 cm

nielit2023feb-scientistd geometry quantitative-aptitude

16.16.11 Geometry: NIELIT 2023 Feb Scientist D | Section D | Question: 92



A train enters into a tunnel AB at A and exits at B . A jackal is sitting at O in another by passing tunnel AOB , which is connected to AB at A and B , where OA is perpendicular to OB . A cat is sitting at P inside the tunnel AB making the shortest possible distance between O and P , such that $AO = 30 \text{ km}$ and $PB = 32 \text{ km}$. When a train before entering into the tunnel AB makes a whistle somewhere before A , the jackal and cat run towards A , they meet with accident at the entrance A . The ratio of speeds of jackal and cat is :

A. 2 : 3

B. 4 : 3

C. 5 : 3

D. 3 : 2

nielit2023feb-scientistd analytical-aptitude geometry quantitative-aptitude

16.16.12 Geometry: NIELIT Scientist B 2020 November: 24



If a cube with length, height and width equal to 10 cm , is reduced to a smaller cube of height, length and width of 9 cm then reduction in volume is :

A. 172 cm^3

B. 729 cm^3

C. 271 cm^3

D. None of the options

nielit-scb-2020 quantitative-aptitude geometry

Answer key 

16.17

Installments (1)

16.17.1 Installments: NIELIT 2023 Feb Scientist D | Section D | Question: 95



A mobile phone is available for Rs.19650 on cash payment or for Rs.3100 cash down payment and three equal annual installments. If the shopkeeper charges

interest at the rate of 10% per annum compounded annually. Calculate the amount of each installment.

- A. Rs. 6655 B. Rs. 6555 C. Rs. 6755 D. Rs. 6855

nielit2023feb-scientistd compound-interest quantitative-aptitude installments

16.18

Logarithms (3)

16.18.1 Logarithms: NIELIT 2021 Dec Scientist A - Section A: 35



Which of the following is true ?

- A. $\log_{17} 275 = \log_{19} 375$ B. $\log_{17} 275 > \log_{19} 375$
C. $\log_{17} 275 < \log_{19} 375$ D. None of the these

nielit2021dec-scientista logarithms quantitative-aptitude

16.18.2 Logarithms: NIELIT 2021 Dec Scientist A - Section A: 36



$\log(x + 3) + \log(x + 5) = \log 35$, solve for x :

- A. 1 B. 2 C. 3 D. 4

nielit2021dec-scientista logarithms quantitative-aptitude

Answer key

16.18.3 Logarithms: NIELIT 2021 Dec Scientist B - Section A: 28



$\frac{1}{2} \log_{10} 25 - 2 \log_{10} 3 + \log_{10} 18$ equals :

- A. 18 B. 1 C. $\log_{10} 3$ D. None of these

nielit2021dec-scientistb logarithms quantitative-aptitude

Answer key

16.19

Logical Reasoning (1)

16.19.1 Logical Reasoning: NIELIT 2023 Feb Scientist C | Section A | Question: 9



Four usual dice are thrown on the ground. The total of numbers on the top faces of these four dice is 13 as the top faces showed 4, 3, 1 and 5 respectively. What is the total of the faces touching the ground?

- A. 12 B. 13
C. 15 D. Cannot be determined

nielit2023feb-scientistc probability logical-reasoning

Answer key

16.20

Mixtures (1)

16.20.1 Mixtures: NIELIT 2023 Feb Scientist C | Section C | Question: 90



Three vessels having volumes in the ratio of $1 : 2 : 3$ are full of a mixture of coke and soda. In the first vessel, ratio of coke and soda is $2 : 3$, in second, $3 : 7$ and in third, $1 : 4$. If the liquid in all three vessels were mixed in a bigger container, what is the resulting ratio of coke and soda?

- A. $4 : 11$ B. $5 : 7$ C. $7 : 11$ D. $7 : 5$

nielit2023feb-scientistc ratio-proportion mixtures quantitative-aptitude

16.21

Modular Arithmetic (1)

16.21.1 Modular Arithmetic: NIELIT 2023 Feb Scientist D | Section C | Question: 70



What is the remainder when 153×698 is divided by 5?

- A. 1 B. 2 C. 4 D. 3

nielit2023feb-scientistd modular-arithmetic quantitative-aptitude

Answer key

16.22

Number Series (3)

16.22.1 Number Series: NIELIT 2021 Dec Scientist B - Section A: 4



Directions (3 – 5) : In each of the following questions, one number is wrong in the series. Find out the wrong number.

1, 9, 25, 49, 86, 121 :

- A. 25 B. 121 C. 166 D. 86

nielit2021dec-scientistb number-series analytical-aptitude

Answer key

16.22.2 Number Series: NIELIT 2021 Dec Scientist B - Section A: 5



Directions (3 – 5) : In each of the following questions, one number is wrong in the series. Find out the wrong number.

701, 348, 173, 85, 41, 19, 8 :

- A. 173 B. 41 C. 19 D. 348

nielit2021dec-scientistb number-series analytical-aptitude quantitative-aptitude

Answer key 

16.22.3 Number Series: NIELIT 2022 April Scientist B | Section A | Question: 13

In the following question, a number series is given with one term missing. Choose the correct alternative that will continue the same pattern and fill in the blank spaces.



97, 86, 73, 58, 45, (. . .)

- A. 24 B. 34 C. 44 D. 54

nielit2022apr-scientistb number-series analytical-aptitude

16.23 Number System (5)

16.23.1 Number System: NIELIT 2018-2



Divide 88 into four parts such as first part known as a , second part b , third part c , and fourth part is d , when 5 is added to the first part, 5 is subtracted from the second part, 3 is multiplied by the third part and the fourth part is divided by 5, then all results value are equal. Find the value of a , b , c and d respectively.

- A. 7, 17, 4, 60 B. 8, 25, 5, 50 C. 10, 30, 3, 45 D. 17, 7, 4, 60

nielit-2018 quantitative-aptitude number-system

Answer key 

16.23.2 Number System: NIELIT 2021 Dec Scientist A - Section A: 26



The LCM of two numbers is 45 times their HCF. One number is 125 and the sum of their HCF and LCM is 1150. Find the other number.

- A. 275 B. 215 C. 230 D. 225

nielit2021dec-scientista quantitative-aptitude number-system

Answer key 

16.23.3 Number System: NIELIT 2022 April Scientist B | Section A | Question: 16

The total number of digits used in numbering the pages of a book having 366 pages is :



- A. 730 B. 792 C. 990 D. 1098

nielit2022apr-scientistb analytical-aptitude counting number-system

Answer key 

16.23.4 Number System: NIELIT 2022 April Scientist B | Section A | Question: 39

If the number $467X4$ is divisible by 9, find the value of the digit marked as X .



A. 4

B. 5

C. 6

D. 7

nielit2022apr-scientistb quantitative-aptitude number-system

Answer key 

16.23.5 Number System: NIELIT 2023 Feb Scientist C | Section C | Question: 83

Find the two-digit number the quotient of whose division by the product of its digits is equal to $\frac{8}{3}$, and the difference between the required number and the number consisting of the digits written in the reverse order is 18?



A. 86

B. 42

C. 75

D. None of these

nielit2023feb-scientistc analytical-aptitude number-system quantitative-aptitude

16.24

Number Theory (8)

16.24.1 Number Theory: NIELIT 2023 Feb Scientist C | Section C | Question: 74

The Product of three consecutive natural numbers, the first of which is an even number, is always divisible by



A. 12

B. 24

C. 6

D. All of these

nielit2023feb-scientistc quantitative-aptitude number-theory

Answer key 

16.24.2 Number Theory: NIELIT 2023 Feb Scientist C | Section C | Question: 77

Find the number of numbers between 200 and 300, both included, which are not divisible by 2, 3, 4 and 5.



A. 27

B. 26

C. 25

D. 28

nielit2023feb-scientistc counting number-theory permutation-and-combination quantitative-aptitude

16.24.3 Number Theory: NIELIT 2023 Feb Scientist C | Section C | Question: 79

Find the sum of all two digit numbers such that the sum of the digits constituting the number is not less than 7; the sum of the squares of the digits is not greater than 30; the number consisting of the same digits written in the reverse order is not larger than half the given number.



A. 52

B. 51

C. 49

D. 53

16.24.4 Number Theory: NIELIT 2023 Feb Scientist C | Section C | Question: 86

Which one of the following is a composite number?



- A. 101 B. 103 C. 141 D. 137

16.24.5 Number Theory: NIELIT 2023 Feb Scientist C | Section D | Question: 97

Find the number of zeroes in the following multiplication :

$$5 \times 10 \times 15 \times 20 \times 25 \times 30 \times 35 \times 40 \times 45 \times 50$$



- A. 6 B. 7 C. 8 D. 5

16.24.6 Number Theory: NIELIT 2023 Feb Scientist C | Section D | Question: 98

For the positive integers a, b, c, d , find the minimum value of $(-1)^a + (-1)^b + (-1)^c + (-1)^d$, if $a+b+c+d = 1947$.



- A. -1 B. -2 C. -3 D. -4

16.24.7 Number Theory: NIELIT 2023 Feb Scientist D | Section D | Question: 116

Let n be a natural number such that $\frac{1}{2} + \frac{1}{3} + \frac{1}{7} + \frac{1}{n}$ is also a natural number then which of the following statement is true?



- A. 2 divides n B. 3 divides n
C. 7 divides n D. $n > 84$

16.24.8 Number Theory: NIELIT 2023 Feb Scientist D | Section D | Question: 91

The digit in the unit place of the number represented by $7^{95} - 3^{58}$ is :



- A. 0 B. 4 C. 6 D. 7

Answer key 

16.25

Numerical Computation (1)

16.25.1 Numerical Computation: NIELIT 2017 DEC Scientist B - Section B: 9



Find the smallest number y such that $y \times 162$ (y multiplied by 162) is a perfect cube

- A. 24 B. 27 C. 36 D. 38

nielit2017dec-scientistb general-aptitude quantitative-aptitude number-system numerical-computation

Answer key 

16.26

Percentage (14)

16.26.1 Percentage: NIELIT 2023 Feb Scientist C | Section C | Question: 69



Arvind spends 75% of his income and saves the rest. His income is increased by 20% and he increased his expenditure by 10%. Then the increase in savings in percentage is

- A. 55 B. 52 C. 50 D. 48

nielit2023feb-scientistc percentage quantitative-aptitude

Answer key 

16.26.2 Percentage: NIELIT 2023 Feb Scientist C | Section C | Question: 70



Two solutions, the first of which contains 0.8 kg and the second 0.6 kg of salt, were poured together and 10 kg of a new salt solution were obtained. Find the weight of the first and of the second solutions in the mixture if the first solution is known to contain 10 percent more of salt of than the second.

- A. 4 kg, 6 kg B. 3 kg, 7 kg
C. 4 kg, 9 kg D. 5 kg, 9 kg

nielit2023feb-scientistc system-of-equations percentage quantitative-aptitude

16.26.3 Percentage: NIELIT 2023 Feb Scientist C | Section C | Question: 73



If the length of the rectangle is increased by 50% and breadth is decreased by 50% what is the change in the area of the rectangle :

- A. Increase by 12.5% B. Decrease by 12.5%
C. No change D. Decrease by 25%

16.26.4 Percentage: NIELIT 2023 Feb Scientist C | Section D | Question: 106

Find a single discount equivalent to the discount series of 20%, 10%, 5%?

- A. 30% B. 31.6% C. 68.4% D. 35%

16.26.5 Percentage: NIELIT 2023 Feb Scientist C | Section D | Question: 110

8% of the people eligible to vote are between 20 and 25 years of age. In an election 85% of those eligible to vote, who were between 20 and 25, actually voted. In that election the number of persons between 20 and 25, who actually voted, was what percentage of those eligible to vote?

- A. 4.2 B. 6.4 C. 6.8 D. 8

16.26.6 Percentage: NIELIT 2023 Feb Scientist C | Section D | Question: 119

When the price of mobile was reduced by 20% the number of mobiles sold is increased by 40%. The effect on the sale was:

- A. 12% increase B. 12% decrease
C. 30% increase D. 40% decrease

16.26.7 Percentage: NIELIT 2023 Feb Scientist D | Section C | Question: 68

At an election, the candidate who got 56% of the votes cast won by 1440 votes. Find the total number of voters on the voting list if 80% people cast their vote and there were no invalid votes.

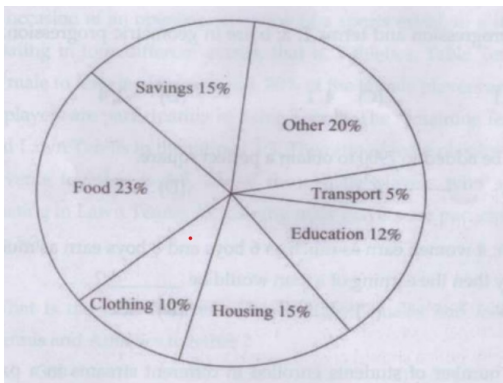
- A. 3600 B. 7200 C. 18000 D. 15000

16.26.8 Percentage: NIELIT 2023 Feb Scientist D | Section C | Question: 78

Directions for **Q 78 to 82**: The circle graph given here shows the spending by a family on various items during the year 1999. Study the graph and answer these

questions.

Percent of money spent by a family on various items during 1999.



If the total expenditure of the family for the year 1999 was Rs. 46000, the family saved during the year:

- A. Rs.1500
- B. Rs.15000
- C. Rs.6900
- D. Rs.3067 approx.

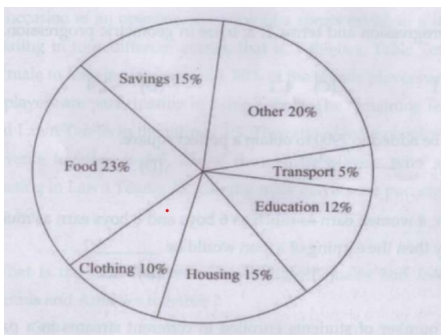
nielit2023feb-scientistd percentage data-interpretation analytical-aptitude quantitative-aptitude

16.26.9 Percentage: NIELIT 2023 Feb Scientist D | Section C | Question: 80



Directions for **Q 78 to 82**: The circle graph given here shows the spending by a family on various items during the year 1999. Study the graph and answer these questions.

Percent of money spent by a family on various items during 1999.



If the total amount spent was Rs. 46000, how much money was spent on clothing and housing together?

- A. Rs. 11500
- B. Rs. 1150
- C. Rs. 10000
- D. Rs. 15000

nielit2023feb-scientistd analytical-aptitude percentage

16.26.10 Percentage: NIELIT 2023 Feb Scientist D | Section D | Question: 114



A retailer bought 20 kg tea at a discount of 10% on market price. Besides 1 kg tea was freely offered to him by the wholesaler at the purchase of 20 kg tea. Now

he sells all the tea at the marked price to a customer. What is the profit percentage of retailer?

- A. 30% B. 12% C. 16.66% D. 10%

nielit2023feb-scientistd quantitative-aptitude percentage

16.26.11 Percentage: NIELIT 2023 Feb Scientist D | Section D | Question: 93



A, B and C went to market to purchase the rings whose costs are same. But each ring is available with two successive discounts. A, B and C avails two successive discounts at the rate of 5% and 20%, 10% and 15%, 12% and 13% respectively. Who gets the maximum discounts?

- A. A B. B
C. C D. All gets the same discount.

nielit2023feb-scientistd percentage analytical-aptitude quantitative-aptitude

16.26.12 Percentage: NIELIT 2023 Feb Scientist D | Section D | Question: 94



The cost of food accounted for 25% of the income of a particular family. If the income gets raised by 20%, then what should be the percentage point decrease in the food expenditure as a percentage of the total income to keep the food expenditure unchanged between two years?

- A. 3.5% B. 8.33% C. 4.16% D. 5%

nielit2023feb-scientistd percentage quantitative-aptitude analytical-aptitude

16.26.13 Percentage: NIELIT Scientific Assistant A 2020 November: 24



If 5% income of P is equal to 15% income of Q and 10% income of Q is equal to 20% income of R. If income of R is ₹2000, then what are the incomes of P and Q respectively?

- A. ₹4000 and ₹8000 B. ₹12000 and ₹4000
C. ₹15000 and ₹5000 D. ₹18000 and ₹6000

nielit-sta-2020 quantitative-aptitude percentage

Answer key

16.26.14 Percentage: NIELIT Scientist B 2020 November: 40



In an office, 30% of the employees were women and 70% of the employees were above the age of 40 years, out of which 60% are men. Find the percentage of women employees who are above 40 years out of the total number of women employees.

A. 96%

B. 93.33%

C. 70.44%

D. 80.66%

nielit-scb-2020 probability percentage analytical-aptitude

Answer key

16.27

Permutation and Combination (2)

16.27.1 Permutation and Combination: NIELIT 2021 Dec Scientist B - Section B: 27

The number of 4 digit numbers which contain not more than two different digits is :



A. 576

B. 567

C. 513

D. 504

nielit-2021-it-dec-scientistb counting permutation-and-combination quantitative-aptitude

16.27.2 Permutation and Combination: NIELIT 2021 Dec Scientist B - Section B: 62

The number of 4 digit numbers which contain not more than two different digits is :



A. 576

B. 567

C. 513

D. 504

nielit2021dec-scientistb counting permutation-and-combination number-system

Answer key

16.28

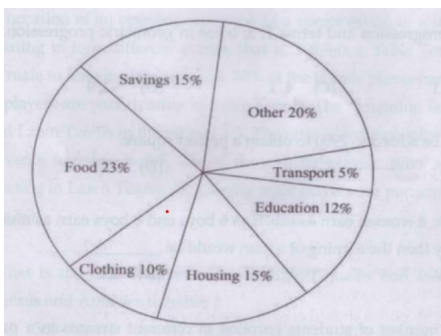
Pie Chart (2)

16.28.1 Pie Chart: NIELIT 2023 Feb Scientist D | Section C | Question: 79



Directions for **Q 78 to 82**: The circle graph given here shows the spending by a family on various items during the year 1999. Study the graph and answer these questions.

Percent of money spent by a family on various items during 1999.



If the total amount spent during the year 1999 was Rs. 46000, the amount spent on food was

A. Rs. 2000

B. Rs. 10580

C. Rs. 23000

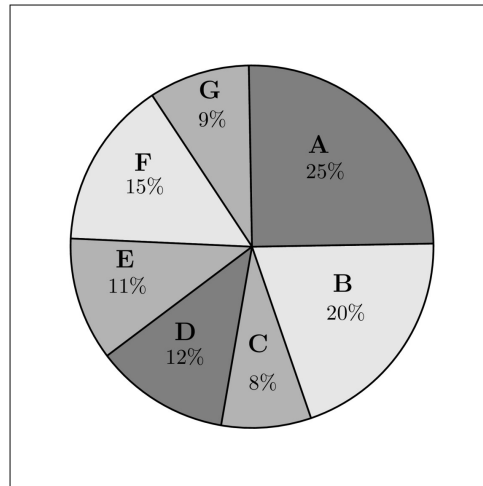
D. Rs. 2300

nielit2023feb-scientistd analytical-aptitude quantitative-aptitude pie-chart data-interpretation

16.28.2 Pie Chart: NIELIT Scientist B 2020 November: 3



Study the following graph and the table and answer the question given below.
(Data of different states regarding population of states in the year 2018)



Total population of the given States = 3276000

States	Sex and Literacy wise Population Ratio			
	Sex		Literacy	
	M	F	Educated	Non-Educated
A	5	3	2	7
B	3	1	1	4
C	2	3	2	1
D	3	5	3	2
E	3	4	4	1
F	3	2	7	2
G	3	4	9	4

If in the year 2018, population of F is increased by 10% and population of B is increased by 12% as compared to the previous year, then _____ is the ratio of populations of F and B in 2017.

- A. 42 : 55 B. 62 : 55 C. 42 : 11 D. 44 : 5

nielit-scb-2020 pie-chart data-interpretation quantitative-aptitude

Answer key

16.29

Pipe Cistern (1)

16.29.1 Pipe Cistern: NIELIT 2023 Feb Scientist D | Section C | Question: 75



A cistern has two taps which fill it in 12 minutes and 15 minutes, respectively. There is also a waste pipe in the cistern. When all the pipes are opened, the empty

cistern is full in 20 minutes. How long will the waste pipe take to empty a full cistern?

- A. 8 minutes B. 10 minutes C. 12 minutes D. 16 minutes

nielit2023feb-scientistd quantitative-aptitude work-time pipe-cistern

Answer key 

16.30

Probability (1)

16.30.1 Probability: NIELIT 2023 Feb Scientist C | Section C | Question: 62



A candidate takes a test and attempts all the 100 questions in it. While any correct answer fetches 1 mark, wrong answers are penalized as follows; one-tenth of the questions carry $1/10$ negative mark each, one-fifth of the questions carry $1/5$ negative marks each and the rest of the questions carry $1/2$ negative mark each. Un-attempted questions carry no marks. What is the difference between the maximum and the minimum marks that he can score?

- A. 100 B. 120
C. 140 D. None of these

nielit2023feb-scientistc probability analytical-aptitude quantitative-aptitude

16.31

Profit Loss (6)

16.31.1 Profit Loss: NIELIT 2018-5



Suppose a fraud shopkeeper sells rice to the customer at the cost price, but he uses a false weight of 900 gm for a kg then his percentage gain is _____

- A. 5.75% B. 5.56% C. 5.20% D. 5.00%

nielit-2018 general-aptitude quantitative-aptitude profit-loss

Answer key 

16.31.2 Profit Loss: NIELIT 2023 Feb Scientist C | Section D | Question: 100



Mahesh and Umesh purchase a radio each for the same price and both mark up their respective radios by the same amount. Mahesh gives a discount of Rs. 20/- followed by another discount of 20% on the reduced price. While Umesh gives a discount of 20% followed by a discount of Rs 20/—. If Mahesh's profit percent is equal to thrice Umesh's loss percent. The profit (in Rs) of Mahesh on his radio is:

- A. Rs. 10 B. Rs. 3 C. Rs. 15 D. Rs. 7

nielit2023feb-scientistc profit-loss quantitative-aptitude analytical-aptitude

16.31.3 Profit Loss: NIELIT 2023 Feb Scientist C | Section D | Question: 93

A dishonest dealer sells the goods at $6\frac{1}{4}\%$ loss on cost price but uses $12\frac{1}{4}\%$ less weight. What is his percentage profit or loss?

- A. $7\frac{1}{4}\%$ B. $4\frac{1}{7}\%$ C. $7\frac{1}{7}\%$ D. $4\frac{1}{4}\%$

nielit2023feb-scientistc percentage profit-loss quantitative-aptitude

Answer key

16.31.4 Profit Loss: NIELIT 2023 Feb Scientist D | Section D | Question: 104

A milkman purchases the milk at Rs. x per litre and sells it at Rs. $2x$ per litre still he mixes 2 litres water with every 6 litres of pure milk. What is the profit percentage?

- A. 116% B. 166.66% C. 60% D. 100%

nielit2023feb-scientistd quantitative-aptitude percentage profit-loss

Answer key

16.31.5 Profit Loss: NIELIT 2023 Feb Scientist D | Section D | Question: 115

A pen manufacturing company produces very fine quality of writing pens. Company knows that on an average 10% of the produced pens are always defective, so are rejected before packing. Company promises to deliver 7200 pens to its wholesaler at Rs 10 each. It estimates the overall profit on all manufactured pens to be 25%. What is the manufacturing cost of each pen?

- A. Rs. 6 B. Rs. 5.6 C. Rs. 8 D. Rs. 7.2

nielit2023feb-scientistd percentage profit-loss quantitative-aptitude

16.31.6 Profit Loss: NIELIT Scientific Assistant A 2020 November: 25

A businessman purchases an item at a certain price and marks its price up by 30%. He sells the item at a certain discount on markup price and makes a net profit of 4% on the whole transaction. Find the discount given by businessman on markup price.

- A. 10 B. 15 C. 26 D. 20

nielit-sta-2020 quantitative-aptitude profit-loss

Answer key

16.32

Quadratic Equations (1)

16.32.1 Quadratic Equations: NIELIT 2018-17



If $2a + 3b + c = 0$, then at least one root of the equation $ax^2 + bx + c = 0$, lies in the interval:

- A. (0,1) B. (1,2) C. (2,3) D. (1,3)

nielit-2018 general-aptitude quantitative-aptitude quadratic-equations

Answer key

16.33

Ratio Proportion (14)

16.33.1 Ratio Proportion: NIELIT 2021 Dec Scientist A - Section A: 23



The length, breadth and height of a room are in the ratio of $3 : 2 : 1$. If its volume be 1296 m^3 , find its breadth.

- A. 12 m B. 18 m C. 16 m D. 24 m

nielit2021dec-scientista quantitative-aptitude ratio-proportion

Answer key

16.33.2 Ratio Proportion: NIELIT 2021 Dec Scientist B - Section A: 22



A factory employs skilled workers, unskilled workers and clerks in the proportion $8 : 5 : 1$, and the wages of a skilled worker, an unskilled worker and a clerk are in the ratio $5 : 2 : 3$. When 20 unskilled workers are employed, the total daily wages of all (skilled workers, unskilled workers and clerks) amount to ₹ 318. The wages paid to each category of workers are :

- A. ₹ 240, ₹ 60, ₹ 18
B. ₹ 200, ₹ 90, ₹ 28
C. ₹ 150, ₹ 108, ₹ 60
D. ₹ 250, ₹ 50, ₹ 18

nielit2021dec-scientistb analytical-aptitude ratio-proportion

Answer key

16.33.3 Ratio Proportion: NIELIT 2021 Dec Scientist B - Section A: 24



In a mixture of 60 L, the ratio of milk and water is $2 : 1$. If the ratio of milk and water is to be $1 : 2$, then the amount of water to be further added must be :

- A. 40 L B. 30 L C. 20 L D. 60 L

nielit2021dec-scientistb ratio-proportion quantitative-aptitude analytical-aptitude

Answer key

16.33.4 Ratio Proportion: NIELIT 2022 April Scientist B | Section A | Question: 2

Ramesh and Suresh enter into a partnership with capitals in the ratio of 10 : 12. At the end of 8 months, Ramesh withdraws. If they receive profits in the ratio of 10 : 18. Find how long Suresh's capital was used.



- A. 7 months B. 8 months C. 10 months D. 12 months

nielit2022apr-scientistb quantitative-aptitude ratio-proportion

Answer key

16.33.5 Ratio Proportion: NIELIT 2022 April Scientist B | Section A | Question: 9

Gold is 19 times as heavy as water and copper 9 times as heavy as water. In what ratio should these metal be mixed so that the mixture may be 15 times as heavy as water ?



- A. 3 : 2 B. 9 : 5 C. 2 : 3 D. 7 : 5

nielit2022apr-scientistb ratio-proportion quantitative-aptitude analytical-aptitude

Answer key

16.33.6 Ratio Proportion: NIELIT 2023 Feb Scientist C | Section A | Question: 15

A box contains two dozen of glasses. If it is dropped on a concrete floor then which of the following cannot be the ratio of broken and unbroken glasses.



- A. 2 : 1 B. 5 : 1 C. 6 : 1 D. 3 : 1

nielit2023feb-scientistc probability counting ratio-proportion analytical-aptitude

16.33.7 Ratio Proportion: NIELIT 2023 Feb Scientist C | Section C | Question: 63

A mixture of cement, sand and gravel in the ratio of 1 : 2 : 4 by volume is required. A person wishes to measure out quantities by weight. He finds that the weight of one cubic foot of cement is 94 kg, of sand 100 kg and gravel 110 kg. What should be the ratio of cement, sand and gravel by weight in order to give a proper mixture?



- A. 47 : 100 : 220 B. 94 : 100 : 220
C. 47 : 200 : 440 D. None of these

nielit2023feb-scientistc ratio-proportion quantitative-aptitude

16.33.8 Ratio Proportion: NIELIT 2023 Feb Scientist C | Section C | Question: 85

A precious stone weighing 35 grams worth ₹12,500 is accidentally dropped and gets broken into two pieces having weights in the ratio of 2 : 5. If the price varies as the square of the weight then find the loss incurred.



A. ₹ 5750

B. ₹ 6000

C. ₹ 5102

D. ₹ 5000

nielit2023feb-scientistc quantitative-aptitude ratio-proportion analytical-aptitude

16.33.9 Ratio Proportion: NIELIT 2023 Feb Scientist D | Section C | Question: 61

The digits of a two-digit number are in the ratio of $2 : 3$ and the number obtained by interchanging the digits is bigger than the original number by 27. What is the original number?



A. 63

B. 48

C. 96

D. 69

nielit2023feb-scientistd analytical-aptitude ratio-proportion number-system quantitative-aptitude

16.33.10 Ratio Proportion: NIELIT 2023 Feb Scientist D | Section C | Question: 64

Directions for (64 and 65): Read the following information carefully to answer the following questions. On the occasion of an opening ceremony of a sports event, in a stadium, there are 600 players who are participating in four different events, that is, Athletics, Table Tennis, Kho-Kho and Lawn Tennis. The ratio of male to female players is $11 : 4$. 30% of the female players are participating in Athletics. 10% of the female players are participating in Table Tennis. The remaining female players are participating in KhoKho and Lawn Tennis in the ratio of $1 : 3$. The ratio of male players who are participating in Athletics and other events together is $3 : 5$. 4% of those male players who are not participating in Athletics are participating in Lawn Tennis. Remaining male players are participating in Table Tennis and Kho-Kho in the ratio $5 : 3$.



What is the total number of players (both males and females together) participating in Table Tennis and Athletics together?

A. 360

B. 358

C. 374

D. None of these

nielit2023feb-scientistd analytical-aptitude ratio-proportion percentage

16.33.11 Ratio Proportion: NIELIT 2023 Feb Scientist D | Section C | Question: 65

Directions for (64 and 65): Read the following information carefully to answer the following questions. On the occasion of an opening ceremony of a sports event, in a stadium, there are 600 players who are participating in four different events, that is, Athletics, Table Tennis, Kho-Kho and Lawn Tennis. The ratio of male to female players is $11 : 4$. 30% of the female players are participating in Athletics. 10% of the female players are participating in Table Tennis. The remaining female players are participating in KhoKho and Lawn Tennis in the ratio of $1 : 3$. The ratio of male players who are participating in Athletics and other events together is $3 : 5$. 4% of those male players who are not participating in Athletics are participating in Lawn Tennis. Remaining male players are participating in Table Tennis and Kho-Kho in the ratio $5 : 3$.



What is the ratio of the male players participating in Lawn Tennis to the female players participating in Table Tennis?

- A. 11 : 72
C. 11 : 16
B. 11 : 38
D. None of these

nielit2023feb-scientistd analytical-aptitude ratio-proportion percentage

16.33.12 Ratio Proportion: NIELIT 2023 Feb Scientist D | Section C | Question: 67

Two numbers are in the ratio $A : B$. When 1 is added to both the numerator and the denominator, the ratio gets changed to C / D . Again, when 1 is added to both the numerator and the denominator, it becomes $1 / 2$. Find the sum of A and B .



- A. 3
B. 4
C. 5
D. 6

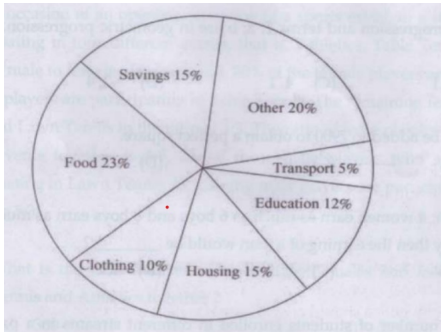
nielit2023feb-scientistd ratio-proportion algebra analytical-aptitude quantitative-aptitude

16.33.13 Ratio Proportion: NIELIT 2023 Feb Scientist D | Section C | Question: 82

Directions for **Q 78 to 82**: The circle graph given here shows the spending by a family on various items during the year 1999. Study the graph and answer these questions.



Percent of money spent by a family on various items during 1999.



The ratio of the total amount of money spent on housing to that spent on education was:

- A. 5 : 2
B. 2 : 5
C. 4 : 5
D. 5 : 4

nielit2023feb-scientistd analytical-aptitude data-interpretation percentage ratio-proportion

16.33.14 Ratio Proportion: NIELIT 2023 Feb Scientist D | Section D | Question: 102

A cat takes 5 leaps for every 4 leaps of a dog, but 3 leaps of the dog are equal to 4 leaps of the cat. What is the ratio of the speed of the cat to that of dog?



- A. 16 : 15
B. 15 : 11
C. 11 : 15
D. 15 : 16

Answer key 

16.34

Remainder (1)

16.34.1 Remainder: NIELIT 2023 Feb Scientist C | Section C | Question: 61



What is the greatest number of 4 digits that when divided by any of the numbers 6, 9, 12, 17 leaves a remainder of 1?

- A. 9997 B. 9793 C. 9895 D. 9487

16.35

Sequence Series (2)

16.35.1 Sequence Series: NIELIT 2016 MAR Scientist C - Section B: 16



Evaluate the sum $S = 1 + 1 + \frac{3}{2^2} + \frac{3}{2^3} + \frac{5}{2^4} + \dots$

- A. 1 B. 2 C. 3 D. 4

Answer key 

16.35.2 Sequence Series: NIELIT 2023 Feb Scientist C | Section C | Question: 82



Fourth term of an arithmetic progression is 8. What is the sum of the first 7 terms of the arithmetic progression?

- A. 7 B. 64
C. 56 D. Cannot be determined

Answer key 

16.36

Set Theory (4)

16.36.1 Set Theory: NIELIT 2022 April Scientist B | Section A | Question: 6



The following question is based on the information given below:

Data of 450 candidates, who took part in an examination in social sciences, Mathematics and Science is given as below :

Number of Students who :

- passed in all subjects are 167

- failed in science = 191
- failed in Mathematics = 199
- failed in Social sciences = 175
- failed in all subjects = 60
- passed in Science only = 52
- passed in Mathematics only = 48
- passed in Social sciences only = 62.

How many failed in two subjects only?

- A. 162 B. 152 C. 100 D. 52

nielit2022apr-scientistb analytical-aptitude set-theory quantitative-aptitude

Answer key 

16.36.2 Set Theory: NIELIT 2023 Feb Scientist D | Section C | Question: 63



In an examination, 30% and 35% students respectively failed in history and geography while 27% students failed in both the subjects. If the number of students passing the examination is 248, find the total number of students who appeared in the examination.

- A. 425 B. 380
C. 400 D. None of these

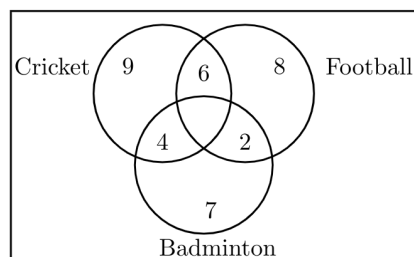
nielit2023feb-scientistd quantitative-aptitude set-theory

Answer key 

16.36.3 Set Theory: NIELIT Scientist B 2020 November: 32



In a class there are 40 students who play at least one game out of Football, Cricket and Badminton.



What percentage of students play only one game?

- A. 50% B. 60% C. 65% D. 70%

nielit-scb-2020 set-theory quantitative-aptitude counting

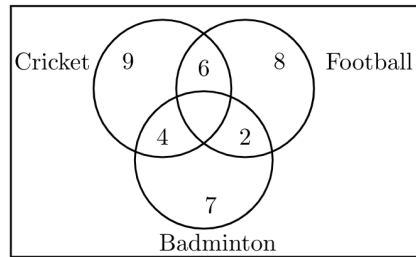
Answer key 

16.36.4 Set Theory: NIELIT Scientist B 2020 November: 33



In a class there are 40 students who play at least one game out of Football,

Cricket and Badminton.



What percentage of students play all the three games?

- A. 4% B. 5% C. 8% D. 10%

nielit-scb-2020 set-theory counting probability quantitative-aptitude

Answer key

16.37 Simple Compound Interest (1)

16.37.1 Simple Compound Interest: NIELIT Scientific Assistant A 2020 November: 32

₹ 1000 doubled in 10 years when compounded annually. How many more years will it take to get another ₹ 2000 compound interest?



- A. 5 years B. 10 years C. 3 years D. 4 years

nielit-sta-2020 quantitative-aptitude simple-compound-interest

Answer key

16.38 Simple Interest (1)

16.38.1 Simple Interest: NIELIT 2021 Dec Scientist A - Section A: 22

A certain sum of money amounts to ₹ 6600 in 4 years at a certain rate percent simple interest. If the rate of interest be increased by its 25%, the same sum would amount to ₹ 7000 during the same period. Find the sum.



- A. ₹ 6000 B. ₹ 5500 C. ₹ 5000 D. ₹ 7000

nielit2021dec-scientista quantitative-aptitude simple-interest

16.39 Speed Time Distance (6)

16.39.1 Speed Time Distance: NIELIT 2022 April Scientist B | Section A | Question: 19

A train moving at 50 km/hr crosses a bridge in 45 seconds. The length of train is 150 meters. Find the length of the bridge.



- A. 525 m B. 545 m C. 475 m D. 575 m

nielit2022apr-scientistb speed-time-distance quantitative-aptitude

Answer key

16.39.2 Speed Time Distance: NIELIT 2022 April Scientist B | Section A | Question: 5

A man rows downstream at 20 km/hr and rows upstream at 15 km/hr. At what speed he can row in still water ?



- A. 17.5 km/hr B. 18 km/hr C. 20.5 km/hr D. 22 km/hr

nielit2022apr-scientistb quantitative-aptitude speed-time-distance

Answer key 

16.39.3 Speed Time Distance: NIELIT 2023 Feb Scientist C | Section C | Question: 84

In a 100 m race, Shyam runs at 1.66 m/s. If Shyam gives Sujit a start of 4 m and still beats him by 12 seconds, what is Sujit's speed?



- A. 1.11 m/s B. 0.75 m/s
C. 1.33 m/s D. 1 km/h

nielit2023feb-scientistc quantitative-aptitude speed-time-distance

16.39.4 Speed Time Distance: NIELIT 2023 Feb Scientist C | Section D | Question: 92

Two trains going on a parallel line in opposite directions take 10 seconds to cross each other. But if they are going in the same direction the longer train crosses the shorter train in 30 seconds. If the length of the longer train decreased by 50%, the time taken to cross the shorter train while going in the same direction decreases by 8 seconds. The time taken by the longer train to cross a tunnel twice its length, if the difference between the length of the trains is 25 m is :



- A. 30sec B. 24sec
C. 36sec D. 40sec

nielit2023feb-scientistc quantitative-aptitude speed-time-distance

16.39.5 Speed Time Distance: NIELIT 2023 Feb Scientist D | Section C | Question: 90

In a 500 — m race, P runs at 10 m/s. If P gives Q a start of 50 m and still wins the race by 5 seconds, find the speed of Q.



- A. 8 m/s B. 8.18 m/s
C. 9.01 m/s D. 9.19 m/s

nielit2023feb-scientistd quantitative-aptitude speed-time-distance

16.39.6 Speed Time Distance: NIELIT Scientist B 2020 November: 36

Arjun by car takes double the time taken by bus to travel from Delhi to Agra. What is the Speed of the Bus if the Speed of the Car is 40 km/hr?

- A. 40 km/hr B. 60 km/hr C. 80 km/hr D. 30 km/hr

nielit-scb-2020 quantitative-aptitude speed-time-distance

Answer key 

16.40

Statistics (2)

16.40.1 Statistics: NIELIT Scientific Assistant A 2020 November: 61



Find the mode of the following data:

Age	0-6	6-12	12-18	18-24	24-30	30-36	36-42
Frequency	6	11	25	35	18	12	6

- A. 20.22 B. 19.47 C. 21.12 D. 20.14

nielit-sta-2020 quantitative-aptitude statistics

Answer key 

16.40.2 Statistics: NIELIT Scientist B 2020 November: 41



Consists of a question and two statements, labelled (1) and (2), in which certain data are given. You have to decide whether the data given in the statements are sufficient for answering the question. Using the data given in the statements, plus your knowledge of mathematics and everyday facts.

What is the Standard Deviation (SD) of the four numbers A, B, C, D?

Statement (1)

The sum of A, B, C and D is 24.

Statement (2)

The sum of the squares of A, B, C and D is 224.

- A. Only Statement (1) is required for answering the question
B. Only Statement (2) is required for answering the question
C. Both statement together are required to answer the question
D. Answer cannot be ascertained with the given information

nielit-scb-2020 statistics analytical-aptitude

Answer key 

16.41

System of Equations (7)

16.41.1 System of Equations: NIELIT 2021 Dec Scientist A - Section A: 33



If the numerator of a fraction is increased by 2 and the denominator is decreased by 1, then it becomes $\frac{2}{3}$. If the numerator is increased by 1 and the denominator is increased by 2, then it becomes $\frac{1}{3}$. Find the fraction.

A. $\frac{2}{9}$ B. $\frac{2}{7}$ C. $\frac{1}{6}$ D. $\frac{1}{5}$

nielit2021dec-scientista analytical-aptitude system-of-equations quantitative-aptitude

Answer key **16.41.2 System of Equations: NIELIT 2021 Dec Scientist B - Section A: 29**

If $x = p, y = q$ then which of following are p and q respectively for pair of equations $3x - 7y + 10 = 0$ and $y - 2x - 3 = 0$:

A. $-1, 1$ B. $1, 1$ C. $1, 0$ D. $0, 1$

nielit2021dec-scientistb system-of-equations analytical-aptitude

Answer key **16.41.3 System of Equations: NIELIT 2022 April Scientist B | Section A | Question: 35**

Four years ago a man was 6 times as old as his son. After 16 years he will be twice as old as his son. What is the present age of man and his son?

A. 34, 9

B. 33, 7

C. 35, 5

D. 36, 6

nielit2022apr-scientistb system-of-equations analytical-aptitude quantitative-aptitude

Answer key **16.41.4 System of Equations: NIELIT 2022 April Scientist B | Section A | Question: 41**

Sum of present age of Suresh and Dinesh is equal to the age of Hema six years back. Five years from now, the ratio of age of Suresh and Dinesh will be 3 : 2. Rajesh is 5 years older than Hema and his present age is four times the present age of Suresh. What is present age of the Dinesh ?

A. 1 year

B. 10 years

C. 2 years

D. 12 years

nielit2022apr-scientistb analytical-aptitude system-of-equations quantitative-aptitude

Answer key **16.41.5 System of Equations: NIELIT 2023 Feb Scientist C | Section C | Question: 80**

The cost of four mangoes, six guavas and sixteen watermelons is ₹500, while the cost of seven mangoes, nine guavas and nineteen watermelons is ₹620. What is the cost of one mango, one guava and one watermelon?

A. 120

B. 40

C. 150

D. Cannot be determined

nielit2023feb-scientistc quantitative-aptitude system-of-equations

16.41.6 System of Equations: NIELIT 2023 Feb Scientist C | Section D | Question: 112

A boat covers 32 km upstream and 36 km downstream in 7hrs. Also, it covers 40 km

upstream and 48 km downstream in 9 hrs. Find the speed of the stream.



- A. 2 km/hr
C. 1 km/hr

- B. 3 km/hr
D. 5 km/hr

nielit2023feb-scientistc quantitative-aptitude system-of-equations

16.41.7 System of Equations: NIELIT 2023 Feb Scientist C | Section D | Question: 91

One fourth of a herd of camels was seen in forest. Twice the square root of the herd had gone to mountains and remaining 15 camels were seen on the bank of a river. Find the total number of camels.



- A. 24 B. 30 C. 36 D. 42

nielit2023feb-scientistc analytical-aptitude system-of-equations quantitative-aptitude

16.42 Tabular Data (3)

16.42.1 Tabular Data: NIELIT 2021 Dec Scientist B - Section A: 33



Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

YEAR	RURAL		SEMI-URBAN		STATE CAPITAL		METROPOLITAN	
	App	Pass	App	Pass	App	Pass	App	Pass
2015	1652	208	7894	2513	5054	1468	9538	3214
2016	1839	317	8562	2933	7164	3248	10158	4018
2017	2153	932	8139	2468	8258	3159	9695	3038
2018	5032	1798	9432	3528	8529	3628	11247	5158
2019	4915	1668	9784	4015	9015	4311	12518	6328
2020	5628	2392	9969	4263	10725	4526	13624	6449

[*App - Appeared

*Pass - Passed]

What approximate value was the percentage drop in the number of semi-urban candidates appeared from 2016 to 2017 ?

- A. 15 B. 10 C. 5 D. 8

nielit2021dec-scientistb tabular-data percentage data-interpretation quantitative-aptitude

Answer key

16.42.2 Tabular Data: NIELIT 2021 Dec Scientist B - Section A: 34



Read the information given below and on the basis of the information, select the correct alternative for each question given after the information.

YEAR	RURAL		SEMI-URBAN		STATE CAPITAL		METROPOLITAN	
	App	Pass	App	Pass	App	Pass	App	Pass
2015	1652	208	7894	2513	5054	1468	9538	3214
2016	1839	317	8562	2933	7164	3248	10158	4018
2017	2153	932	8139	2468	8258	3159	9695	3038
2018	5032	1798	9432	3528	8529	3628	11247	5158
2019	4915	1668	9784	4015	9015	4311	12518	6328
2020	5628	2392	9969	4263	10725	4526	13624	6449

[*App - Appeared

*Pass - Passed]

In which of the following years was the percentage passed to appeared candidates from semi-urban area the least ?

A. 2015

B. 2016

C. 2017

D. 2018

nielit2021dec-scientistb tabular-data percentage data-interpretation analytical-aptitude quantitative-aptitude

16.42.3 Tabular Data: NIELIT 2023 Feb Scientist D | Section C | Question: 74



The following table shows the number of students enrolled in different streams in a particular college.

SCIENCE		HUMANITIES		COMMERCE		VOCATIONAL	
Boys	Girls	Boys	Girls	Boys	Girls	Boys	Girls
32	18	28	45	42	42	13	30

The ratio of the number of girls studying Humanities to the number of girls studying in all other streams is :

A. 1 : 3

B. 2 : 1

C. 1 : 2

D. 3 : 1

nielit2023feb-scientistd data-interpretation tabular-data

Answer key

16.43

Triangles (2)

16.43.1 Triangles: NIELIT 2018-8



Let us consider the length of the side of a square represented by $2y + 3$. The

length of the side of an equilateral triangle is $4y$. If the square and the equilateral triangle have equal perimeter, then what is the value of y ?

- A. 3 B. 4 C. 6 D. 8

nielit-2018 general-aptitude quantitative-aptitude geometry triangles

Answer key 

16.43.2 Triangles: NIELIT 2023 Feb Scientist C | Section D | Question: 103



The base of a triangle is 15 cm. Two lines are drawn parallel to the base, terminating in the other two sides, and dividing the triangle into three equal areas. The length of the parallel line close to the base is :

- A. $5\sqrt{6}$ cm B. $4\sqrt{3}$ cm
C. 10 cm D. 5.5 cm

nielit2023feb-scientistc geometry area triangles quantitative-aptitude

16.44 Trigonometry (2)

16.44.1 Trigonometry: NIELIT 2023 Feb Scientist C | Section D | Question: 96



The value of the expression $\sqrt{3} \operatorname{cosec} 20^\circ - \sec 20^\circ$ is

- A. 2 B. $2 \frac{\sin 20^\circ}{\sin 40^\circ}$
C. 4 D. $4 \frac{\sin 20^\circ}{\sin 40^\circ}$

nielit2023feb-scientistc trigonometry analytical-aptitude

16.44.2 Trigonometry: NIELIT 2023 Feb Scientist D | Section D | Question: 96



If $\sin x + \sin^2 x = 1$ then $\cos^8 x + 2\cos^6 x + \cos^4 x$ equals to :

- A. 0 B. -1 C. 1 D. 2

nielit2023feb-scientistd quantitative-aptitude trigonometry

Answer key 

16.45 Venn Diagram (6)

16.45.1 Venn Diagram: NIELIT Scientific Assistant A 2020 November: 14



Answer the question on the basis of the following information provided:

The students of a school participate in various sports activities, the distribution of the same is given below:

- Football– 17%

- Handball– 26%
- Badminton– 16%
- Table Tennis– 22%
- Basketball– 19%

Total number of students in the school are 800.

What is the number of girls who take part in handball, if the ratio of boys to girls is 3 : 10 respectively?

- A. 48 B. 80 C. 78 D. 160

nielit-sta-2020 quantitative-aptitude venn-diagram

Answer key 

16.45.2 Venn Diagram: NIELIT Scientific Assistant A 2020 November: 15



Answer the question on the basic of the following information provided:

The students of a school participates in various sports activities, the distribution of the same is given below:

- Football– 17%
- Handball– 26%
- Badminton– 16%
- Table Tennis– 22%
- Basketball– 19%

Total number of students in the school are 800.

What is the respective ratio between the total number of students taking part in Badminton and Table Tennis together and those participating in Basketball and Football together?

- A. 11 : 13 B. 18 : 19 C. 19 : 18 D. 29 : 28

nielit-sta-2020 quantitative-aptitude venn-diagram

Answer key 

16.45.3 Venn Diagram: NIELIT Scientific Assistant A 2020 November: 16



Answer the question on the basic of the following information provided:

The students of a school participates in various sports activities, the distribution of the same is given below:

- Football– 17%
- Handball– 26%
- Badminton– 16%
- Table Tennis– 22%
- Basketball– 19%

Total number of students in the school are 800.

What is the approximate average of number of participants in Handball, Badminton and Basketball?

- A. 162 B. 163 C. 104 D. 169

nielit-sta-2020 quantitative-aptitude venn-diagram

Answer key 

16.45.4 Venn Diagram: NIELIT Scientific Assistant A 2020 November: 17



Answer the question on the basis of the following information provided:

The students of a school participate in various sports activities, the distribution of the same is given below:

- Football– 17%
- Handball– 26%
- Badminton– 16%
- Table Tennis– 22%
- Basketball– 19%

Total number of students in the school are 800.

The number of students taking part in Basketball is approximately what percent more than those taking part in Football?

- A. 10.84% B. 9.92% C. 9.32% D. None of the options

nielit-sta-2020 quantitative-aptitude venn-diagram

Answer key 

16.45.5 Venn Diagram: NIELIT Scientific Assistant A 2020 November: 18



Answer the question on the basis of the following information provided:

The students of a school participate in various sports activities, the distribution of the same is given below:

- Football– 17%
- Handball– 26%
- Badminton– 16%
- Table Tennis– 22%
- Basketball– 19%

Total number of students in the school are 800.

If out of the number of students in Basketball, 69 are girls, what is the difference between the number of boys and girls taking part in Basketball?

- A. 17 B. 23 C. 86 D. 14

nielit-sta-2020 quantitative-aptitude venn-diagram

Answer key 

16.45.6 Venn Diagram: NIELIT Scientific Assistant A 2020 November: 31



A Class has 100 students with roll number from 101 to 200. All the even numbered students study Physics, whose roll number are divisible by 5 study Chemistry & students with roll numbers divisible by 7 study Biology. How many students do not study any of the given subject Physics, Chemistry or Biology?

- A. 35 B. 45 C. 51 D. 62

nielit-sta-2020 quantitative-aptitude venn-diagram

Answer key 

16.46

Work Time (12)

16.46.1 Work Time: NIELIT 2018-3



A software engineer has the capability of thinking 200 lines of code in five minutes and can type 200 lines of code in 10 minutes. He takes a break for five minutes after every 10 minutes. How many lines of code will he complete typing after an hour?

- A. 200 B. 300 C. 400 D. 500

nielit-2018 general-aptitude quantitative-aptitude work-time

Answer key 

16.46.2 Work Time: NIELIT 2018-6



A construction company ready to finish a construction work in 180 days, hired 80 workers each working 8 hours daily. After 90 days, only $\frac{2}{7}$ of the work was completed. How many workers are to be increased to complete the work on time?

Note: If additionally acquired workers do agree to work for 10 hours daily.

- A. 90 workers B. 80 workers C. 65 workers D. 85 workers

nielit-2018 general-aptitude quantitative-aptitude work-time

Answer key 

16.46.3 Work Time: NIELIT 2021 Dec Scientist A - Section A: 24



Abha can do some work in 10 days, Billu can do it in 20 days and Chintu can do it in 40 days. They start working in turns with Abha starting to work on the first day followed by Billu on the second day and by Chintu on the third day and again by Abha on the fourth day and so on, till the work is completed fully. Find the time taken (approx.) to complete the work fully.

- A. 16 days B. 15 days C. 17 days D. 20 days

nielit2021dec-scientista work-time quantitative-aptitude

Answer key 

16.46.4 Work Time: NIELIT 2021 Dec Scientist A - Section A: 29

In a company ABC Ltd. a certain number of engineers can develop a design in 40 days. If there were 5 more engineers, it could be finished in 10 days less. How many engineers were there in the beginning ?

- A. 18 B. 20 C. 25 D. 15

nielit2021dec-scientista quantitative-aptitude work-time

Answer key

16.46.5 Work Time: NIELIT 2022 April Scientist B | Section A | Question: 11

Three pipes A, B and C can fill a cistern in 8 minutes, 12 minutes and 16 minutes respectively. What is the time taken by three pipes to fill the cistern when they are opened together?

- A. 3.7 minutes B. 6 minutes C. 4.5 minutes D. 5 minutes

nielit2022apr-scientistb quantitative-aptitude work-time analytical-aptitude

Answer key

16.46.6 Work Time: NIELIT 2022 April Scientist B | Section A | Question: 36

In a race of 600 m, A can beat B by 50 m, and in another race of 500 m, B can beat C by 60 m. Then by how many meters will A beat C in a race of 400 meters ?

- A. $77\frac{1}{3}$ m B. 80 m C. 70 m D. None of these

nielit2022apr-scientistb quantitative-aptitude work-time

Answer key

16.46.7 Work Time: NIELIT 2023 Feb Scientist C | Section D | Question: 113

A contractor undertakes to build a wall in 50 days. He employs 50 people for the same. However, after 25 days he finds that the work is only 40% complete. How many more men need to be employed to complete the work 10 days before time.

- A. 50 men B. 75 men C. 100 men D. 125 men

nielit2023feb-scientistc quantitative-aptitude work-time

16.46.8 Work Time: NIELIT 2023 Feb Scientist D | Section C | Question: 73

3 men earn as much as 4 women; 4 women earn as much as 6 boys and 8 boys earn as much as 10 girls. If a girl earns Rs. 50/— a day then the earning of a man would be?

A. 115

B. 135

C. 125

D. 150

nielit2023feb-scientistd quantitative-aptitude work-time

Answer key 

16.46.9 Work Time: NIELIT 2023 Feb Scientist D | Section D | Question: 103



Subhash can copy 50 pages in 10hrs,, Subhash and Prakash together can copy 300 pages in 40hrs. In how much time can Prakash copy 30 pages.

A. 13hrs

B. 12hrs

C. 11hrs

D. 9hrs

nielit2023feb-scientistd quantitative-aptitude work-time

Answer key 

16.46.10 Work Time: NIELIT 2023 Feb Scientist D | Section D | Question: 99



A contractor undertakes to build a wall in 50 days. He employs 50 people for the same. However, after 25 days he finds that the work is only 40% complete. How many more men need to be employed to complete the work in time?

A. 50 men

B. 75 men

C. 100 men

D. 25 men

nielit2023feb-scientistd work-time quantitative-aptitude

Answer key 

16.46.11 Work Time: NIELIT Scientific Assistant A 2020 November: 33



Ram can do a piece of work in 5 days, and Sham can do the same in 10 days. With the help of Karan, they finished the work in 2 days. How many days would it take Karan to do the work?

A. 5 days

B. 10 days

C. 15 days

D. 20 days

nielit-sta-2020 quantitative-aptitude work-time

Answer key 

16.46.12 Work Time: NIELIT Scientist B 2020 November: 34



Two taps A and B can fill a tank in 12 minutes and 15 minutes respectively. The tank can be emptied by a third tap C in 6 minutes. If A and B are kept open for 5 minutes in the beginning and then C is opened along with A and B being kept open, the time taken to empty the tank is:

A. 60 minutes

B. 45 minutes

C. 30 minutes

D. 75 minutes

nielit-scb-2020 quantitative-aptitude work-time

Answer key 

Answer Keys

16.0.1	B	16.0.2	A	16.0.3	B	16.0.4	A	16.0.5	B
16.0.6	A	16.0.7	B	16.0.8	C	16.0.9	D	16.0.10	C
16.0.11	TBA	16.0.12	C	16.0.13	A	16.0.14	C	16.0.15	C
16.0.16	D	16.0.17	D	16.0.18	C	16.0.19	C	16.0.20	B
16.0.21	A	16.0.22	A	16.0.23	A	16.0.24	C	16.0.25	A
16.1.1	A	16.2.1	C	16.3.1	D	16.4.1	A	16.5.1	C
16.6.1	A	16.6.2	B	16.6.3	B	16.6.4	C	16.7.1	A
16.7.2	C	16.7.3	C	16.7.4	A	16.7.5	C	16.8.1	C
16.9.1	B	16.9.2	D	16.9.3	D	16.9.4	B	16.10.1	C
16.11.1	D	16.11.2	B	16.12.1	C	16.13.1	D	16.14.1	A
16.14.2	A	16.15.1	C	16.16.1	A	16.16.2	A	16.16.3	D
16.16.4	D	16.16.5	B	16.16.6	A	16.16.7	C	16.16.8	A
16.16.9	A	16.16.10	C	16.16.11	C	16.16.12	C	16.17.1	A
16.18.1	C	16.18.2	B	16.18.3	B	16.19.1	C	16.20.1	A
16.21.1	C	16.22.1	D	16.22.2	D	16.22.3	B	16.23.1	A
16.23.2	B	16.23.3	C	16.23.4	C	16.23.5	D	16.24.1	D
16.24.2	B	16.24.3	A	16.24.4	C	16.24.5	C	16.24.6	B
16.24.7	TBA	16.24.8	B	16.25.1	C	16.26.1	C	16.26.2	A
16.26.3	D	16.26.4	B	16.26.5	C	16.26.6	A	16.26.7	D
16.26.8	C	16.26.9	A	16.26.10	C	16.26.11	A	16.26.12	C
16.26.13	B	16.26.14	B	16.27.1	A	16.27.2	A	16.28.1	B
16.28.2	A	16.29.1	B	16.30.1	A	16.31.1	B	16.31.2	C
16.31.3	TBA	16.31.4	B	16.31.5	A	16.31.6	D	16.32.1	A
16.33.1	A	16.33.2	A	16.33.3	D	16.33.4	D	16.33.5	A
16.33.6	C	16.33.7	A	16.33.8	C	16.33.9	D	16.33.10	D
16.33.11	C	16.33.12	B	16.33.13	D	16.33.14	D	16.34.1	B

16.35.1	D
16.36.4	D
16.39.3	C
16.40.2	C
16.41.5	B
16.42.3	C
16.45.1	D
16.45.6	A
16.46.5	A
16.46.10	D

16.35.2	C
16.37.1	B
16.39.4	B
16.41.1	B
16.41.6	A
16.43.1	A
16.45.2	C
16.46.1	D
16.46.6	D
16.46.11	A

16.36.1	A
16.38.1	C
16.39.5	B
16.41.2	A
16.41.7	C
16.43.2	A
16.45.3	B
16.46.2	B
16.46.7	B
16.46.12	B

16.36.2	C
16.39.1	C
16.39.6	C
16.41.3	A
16.42.1	C
16.44.1	C
16.45.4	D
16.46.3	A
16.46.8	C

16.36.3	B
16.39.2	A
16.40.1	A
16.41.4	A
16.42.2	C
16.44.2	C
16.45.5	D
16.46.4	D
16.46.9	B